

SAMSUNG PRISM

OvaSearch

Multimodal RAG

3rd Edition 2026 – 27

Native C++ • OpenVINO • Local-first retrieval

Theme ID 1: Agentic Code Intelligence

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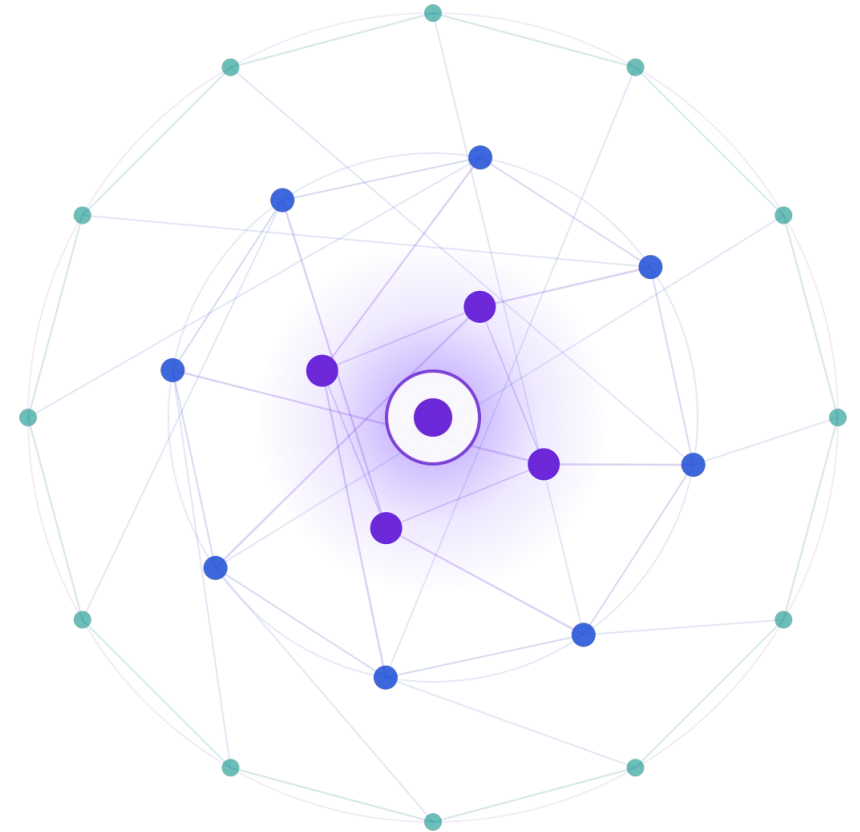
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github.com/lagmator22/OvaSearch



Theme

THEME 1 | AGENTIC CODE
INTELLIGENCE

Find the right code before an agent tries to reason about it.

Challenge goal

- **Rank relevant snippets** for a natural-language query.
- **Optimize retrieval**, generation is outside benchmark scope.
- **Run efficiently** on CPU and rebuild indexes across versions.

How OvaSearch fits

- **Dense retrieval core**: bge embeddings + HNSW vector search.
- **Native runtime**: C++ inference through Intel OpenVINO.
- **Persistent cache**: fast startup and incremental refresh.
- **Multimodal path**: text, documents and images in one index.

NOTE Code-aware chunking (tree-sitter) and the MTEB AppsRetrieval adapter are now built in prism/.

Existing Solutions & Gaps

BASELINE

Keyword / regex

GOOD

Fast and explainable

GAP

Misses semantic intent; brittle naming mismatch

BASELINE

Dense retrievers

GOOD

Semantic matching at scale

GAP

Generic embeddings can blur code structure

BASELINE

LLM reranking

GOOD

Strong reasoning over a shortlist

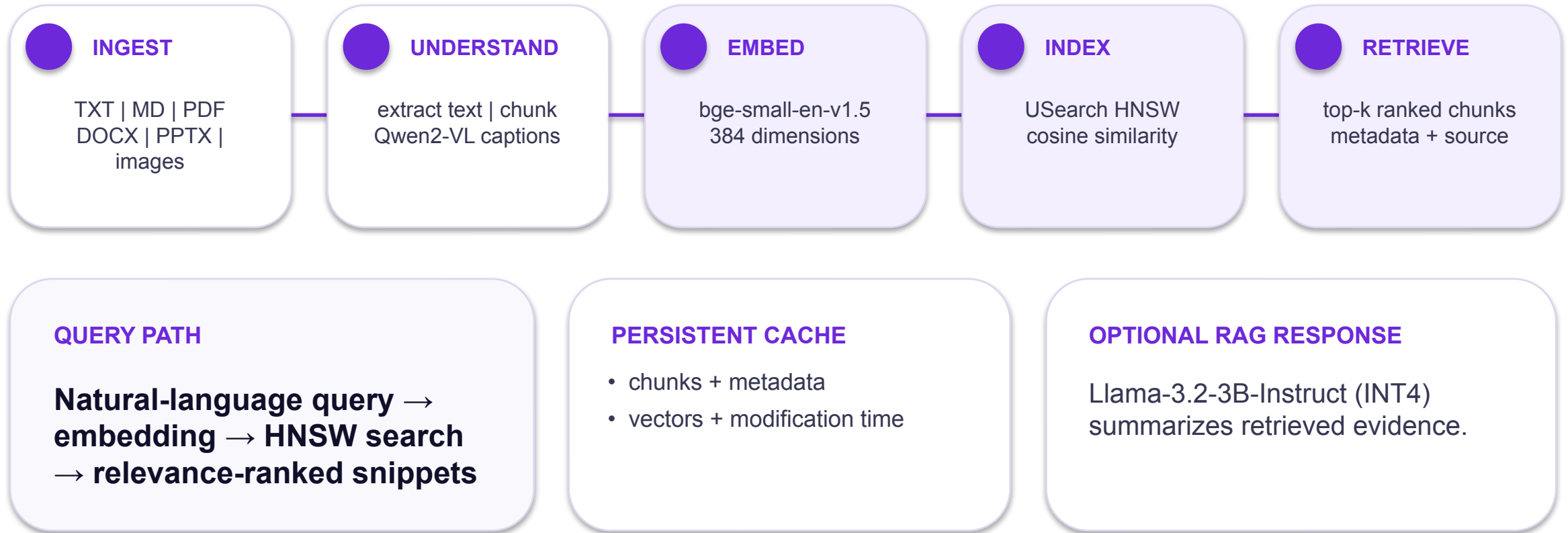
GAP

Slow, costly, and context-limited

Measured on AppsRetrieval (CPU): SFR-Code-400M NDCG@10 0.514, MRR@10 0.470 vs bge-small baseline 0.056 / 0.048.

Our Solution & Architecture

One native pipeline from heterogeneous files to ranked evidence.



Benchmark mode stops after RETRIEVE and exports ranked results, no answer generation.

Demo & Product Walkthrough

TERMINAL | LOCAL EXECUTION

```
$ python -m prism index express --rev 4.21.2  
reused 668 cached vectors, embedded 242
```

```
$ python -m prism search "redirect"  
1. lib/response.js:928-990 res.redirect  
latency: 369 ms on CPU
```

```
$ ./ovasearch  
> What images contain animals?
```

Five-step walkthrough

- | | | |
|----|-----------------|--|
| 01 | Index | prism index: a repo at any git tag or commit |
| 02 | Search | prism search: ranked file:lines + preview |
| 03 | Versions | --all-versions: near-duplicates grouped |
| 04 | Rebuild | New commit: only changed chunks re-embed |
| 05 | C++ app | ./ovasearch for docs and images |

[Demo video: drive.google.com/file/d/1lrPoPA8nCbHOSsjEkVldNYA3m0Zxl8St](https://drive.google.com/file/d/1lrPoPA8nCbHOSsjEkVldNYA3m0Zxl8St)

Tools and Tech Stack Used

NATIVE CORE

C++17

Zero Python runtime overhead during inference

VECTOR SEARCH

USearch

Dense HNSW index with cosine distance

VISION MODEL

Qwen2-VL-2B-Instruct

Image understanding and automatic captions

DOCUMENTS

PDF | DOCX | PPTX

Automatic extraction via utility pipeline

INFERENCE

Intel OpenVINO

CPU/GPU-optimized model execution

TEXT MODEL

bge-small-en-v1.5

384-dimensional semantic embeddings

GENERATION

Llama-3.2-3B INT4

Compact local response generation

BUILD

CMake 3.18+

Auto-detects environment and dependencies

Impact & Use Cases

Private, hardware-accelerated search turns scattered local content into actionable evidence.

01 Developer discovery

Find implementation details, configuration, and behavior across repositories.

02 Enterprise knowledge

Search internal PDFs, decks, docs and images without uploading data to a cloud service.

03 Visual asset search

Ask what appears in diagrams, screenshots and photos through VLM-generated captions.

04 Research synthesis

Retrieve relevant passages quickly, then use a compact local LLM for grounded summaries.

Innovation Highlights, Results & Limitations

INNOVATION

- **Native C++** retrieval and generation runtime
- **One index** for document text and image captions
- **Persistent cache** with file-change invalidation
- **Sliding windows** with configurable overlap
- **INT4 generation** for a smaller local footprint

MEASURED ON CPU*

0.5139

NDCG@10 AppsRetrieval (baseline 0.0564)

0.4696

MRR@10 (baseline 0.0482)

121 s

rebuild Express 4.21.2 (668 of 910 chunks reused)

< 450 ms

prism search latency on CPU

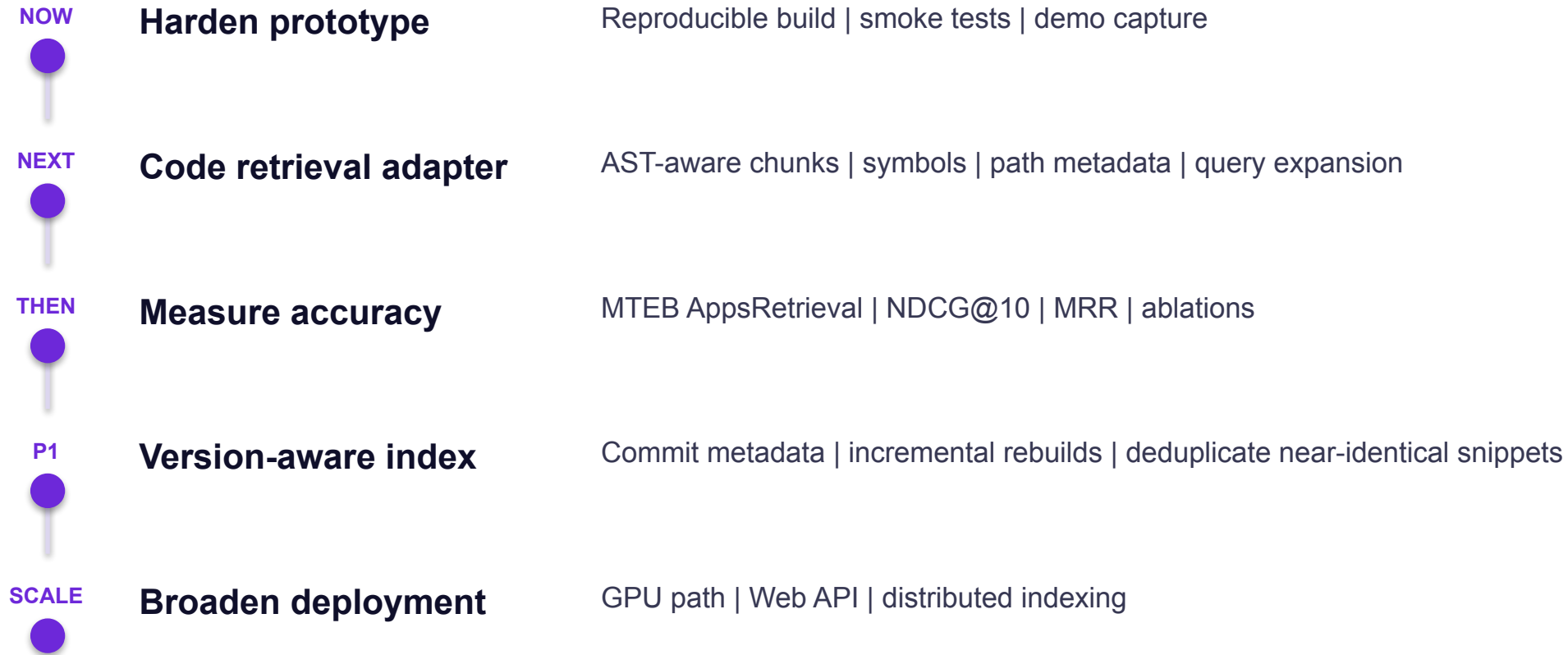
* Apple M2 Pro CPU, SFR-Code-400M.
Details: [prism/eval/RESULTS.md](#)

CURRENT LIMITATIONS

- prism runs the code model on PyTorch CPU; our OpenVINO IR export is not accurate yet.
- BM25 fusion lowered AppsRetrieval, so the benchmark run is dense only.
- AST chunks for Python and JavaScript only; other files use line windows.
- No reranker or query expansion yet.
- Windows not tested; C++ model download still uses Python.

What's Next

Roadmap from strong local RAG engine to benchmark-ready code intelligence.



Brownie Points | Differentiation

OvaSearch pairs a native C++ multimodal engine with a CPU code retriever (prism).

Capability	OvaSearch	Typical Python RAG	Cloud multimodal	Code retriever
Native C++ inference	YES	NO	NO	VARIES
Local/offline pipeline	YES	YES	NO	YES
Text + image retrieval	YES	LIMITED	YES	LIMITED
OpenVINO acceleration	YES	OPTIONAL	NO	OPTIONAL
Persistent cache	YES	VARIES	SERVICE	VARIES
Code benchmark ready	YES	VARIES	NO	YES

Key distinction: local multimodal retrieval plus a measured CPU code retriever with version-aware search.

Submission Checklist | Public GitHub

Repository: github.com/lagmator22/OvaSearch

Working prototype code	YES	Public repository link supplied
README + reproducible setup	YES	Clone, model download, build and usage steps supplied
Demo video ≤ 5 minutes	YES	drive.google.com/file/d/1lrPoPA8nCbmHOSsjEkVldNYA3m0Zxl8St
Presentation file	YES	Completed in supplied template
MTEB AppsRetrieval result JSON	YES	appsretrieval_results.json: NDCG@10 0.5139, MRR@10 0.4696
GitHub Release artifacts	YES	Tag PRISM_GENAI_HACKATHON_Y2026 with JSON, PPTX and PDF attached

[Release: github.com/Lagmator22/OvaSearch/releases/tag/PRISM_GENAI_HACKATHON_Y2026](https://github.com/Lagmator22/OvaSearch/releases/tag/PRISM_GENAI_HACKATHON_Y2026)

OvaSearch

Fast local multimodal retrieval in native C++

TEAM

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github.com/lagmator22/OvaSearch

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India

Thank you